Milestone Project

Enterprise E-Commerce Intelligence & Customer Analytics Platform

This project utilizes the Olist Brazilian E-Commerce Dataset as the underlying data source while simulating a real-world enterprise analytics engagement.

Business Requirement Document (BRD)

1. Project Overview

This milestone project simulates a real-world analytics engagement undertaken by a Data & AI Analytics Team supporting a large-scale e-commerce organization.

Learners are expected to apply concepts covered in Excel, Statistics, MySQL, Power BI, and AI-assisted Analytics to transform raw operational data into meaningful business intelligence and strategic recommendations.

The project is designed to evaluate technical capability, analytical thinking, business understanding, and communication skills.

2. Business Context

A leading e-commerce marketplace operating across multiple regions has experienced rapid growth over the last few years. While order volumes and customer acquisition have increased significantly, leadership teams are facing challenges in understanding the factors driving revenue, customer satisfaction, seller performance, and operational efficiency.

Several departments—including Sales, Customer Experience, Operations, Logistics, and Strategy—currently operate with fragmented reports and limited visibility into key business metrics.

The organization has initiated a strategic analytics program to transform raw operational data into actionable business intelligence and support data-driven decision-making.

As a Data & AI Analyst, you have been assigned to this initiative.

Your responsibility is to analyze the available data assets, identify critical business insights, and develop an executive-level analytics solution that enables leadership teams to monitor performance, identify opportunities, and make informed decisions.

3. Business Problem Statement

Leadership teams currently lack a unified view of business performance across customers, products, sellers, payments, operations, and customer experience.

The organization requires a centralized analytics solution capable of:

- Monitoring overall business performance
- Understanding customer purchasing behaviour
- Evaluating product and seller performance
- Assessing operational efficiency
- Measuring customer satisfaction
- Identifying growth opportunities and business risks
- Supporting strategic decision-making

4. Dataset Acquisition Requirements

The dataset will be provided via a publicly accessible download link shared in the project documentation.

Dataset Source

Olist Brazilian E-Commerce Dataset

Technology Stack

Learners are expected to utilize:

- Microsoft Excel
- MySQL
- Statistical Analysis Techniques
- Power BI
- AI-Assisted Analytics Tools
- GitHub

All learners must use the same dataset version to ensure consistency across evaluation.

5. Available Datasets

Mandatory Datasets

Dataset	Business Purpose
olist_customers_dataset.csv	Customer information
olist_orders_dataset.csv	Order lifecycle and status
olist_order_items_dataset.csv	Product-level transactions
olist_products_dataset.csv	Product catalog
olist_order_payments_dataset.csv	Revenue and payment analysis
olist_order_reviews_dataset.csv	Customer satisfaction analysis
olist_sellers_dataset.csv	Seller performance analysis

Excluded Datasets

The following datasets are intentionally excluded from the project scope:

- olist_geolocation_dataset.csv
- product_category_name_translation.csv

Students are not required to use these datasets.

6. Project Objectives

Develop an end-to-end business intelligence solution capable of delivering meaningful insights into the health and performance of the e-commerce business.

The solution should enable stakeholders to:

- Monitor business performance
- Understand customer behaviour
- Evaluate seller effectiveness
- Improve operational efficiency
- Identify growth opportunities
- Enhance customer experience

- Support strategic decision-making

The project should demonstrate both analytical thinking and business understanding rather than only technical implementation.

7. Scope of Work

Phase 1 – Business & Data Understanding

Students should:

- Understand the business context
- Study each dataset and its purpose
- Identify relationships between datasets
- Develop an analytical approach
- Define key business questions

Deliverables

- Business Understanding Notes
- Dataset Relationship Documentation
- Analytical Approach Summary

Phase 2 – Data Cleaning & Preparation

Students should:

- Assess data quality
- Identify missing values
- Identify inconsistencies
- Validate relationships
- Handle data preparation activities
- Document assumptions and transformations

Students are expected to justify data cleaning decisions based on business impact.

Deliverables

- Cleaned Dataset Files
- Data Quality Assessment Report
- Data Dictionary

Phase 3 – Excel Analysis & Application

Students should perform exploratory analysis using Excel.

Suggested activities include:

- Data profiling
- Trend analysis
- Revenue summaries
- Customer summaries
- Product summaries
- Pivot tables
- Charts and visualizations

Deliverables

- Excel Analysis Workbook

Phase 4 – Database Design & MySQL Analytics

Students should:

- Design an analytical database structure
- Import datasets into MySQL
- Establish relationships
- Create analytical SQL queries

Mandatory Concepts

- Joins
- Aggregations
- Group By
- Filtering
- Subqueries

Deliverables

- SQL Script File
- Database Schema Documentation
- ER Diagram

Phase 5 – Statistical Analysis & Interpretation

Students should apply appropriate statistical methods to investigate business problems.

Examples may include:

- Descriptive statistics
- Distribution analysis
- Trend analysis
- Correlation analysis
- Outlier analysis
- Comparative analysis

Students must justify why a statistical technique was selected and explain how findings support business decisions.

Deliverables

- Statistical Analysis Report

Phase 6 – Power BI Data Modeling & Dashboard Development

Students should develop an executive-ready interactive dashboard.

Mandatory Dashboard Sections

Executive Overview

- Revenue KPIs
- Order KPIs
- Customer KPIs

Customer Intelligence

Revenue Intelligence

Product Intelligence

Seller Intelligence

Customer Experience

Operational Intelligence

Mandatory Expectations

- Data model creation
- Relationship management
- DAX measures
- KPI calculations

- Interactive filtering
- Business storytelling

Deliverables

- Power BI Dashboard (.pbix)
- Dashboard Documentation

Students should design dashboards that support executive decision-making and business performance monitoring rather than focusing solely on visual aesthetics. Dashboards should focus on KPI monitoring, trend analysis, performance diagnostics, and business insights rather than displaying large volumes of raw data.

Phase 7 – AI-Augmented Analytics

Students may leverage AI tools to support:

- Insight generation
- Trend interpretation
- Review analysis
- Business recommendation generation
- Executive summary creation
- Storytelling and presentation preparation

All AI-generated outputs must be validated using evidence from the dataset.

Deliverables

- AI Usage Log
- Prompt Documentation
- Validation Notes
- AI-Assisted Executive Summary

8. Business Questions

The analytical approach is intentionally open-ended.

Students are expected to independently investigate questions related to:

Customer Intelligence

- Customer behaviour
- Customer activity
- Purchase patterns

Revenue Intelligence

- Revenue performance
- Payment behaviour
- Order trends

Product Intelligence

- Product contribution
- Category performance

Seller Intelligence

- Seller effectiveness
- Seller contribution

Customer Experience

- Customer satisfaction
- Review patterns

Operational Intelligence

- Delivery and fulfilment performance
- Order lifecycle efficiency

Students may explore additional business questions beyond those listed above.

8A. Analytical Differentiation Framework (Mandatory)

To ensure uniqueness of student outcomes and avoid repetition in portfolio outputs, each learner is required to apply a differentiated analytical approach while working on the same dataset and core business problem.

Although all students will use the Olist Brazilian E-Commerce Dataset and the same core technology stack, their analysis, interpretation, and recommendations must vary based on assigned analytical perspectives and self-defined business hypotheses.

1. Assigned Analytical Lens (Mandatory Variation)

Each learner must adopt **one primary business perspective** for analysis:

- Customer Retention & Loyalty Analyst
- Revenue Growth Strategist

- Logistics & Delivery Efficiency Analyst
- Seller Performance & Marketplace Health Analyst
- Product & Category Intelligence Analyst
- Customer Experience & Satisfaction Analyst

The selected lens must be clearly stated in the README file.

2. Unique KPI Requirement (Mandatory)

In addition to standard KPIs (Revenue, Orders, Customers), each learner must design and include at least **one unique KPI** relevant to their analytical lens.

Examples include:

- Customer Retention Rate
- Repeat Purchase Ratio
- Delivery Delay Impact Score
- Seller Reliability Index
- Payment Preference Concentration Score
- Customer Risk Score

This KPI must be defined, justified, and implemented in Power BI or SQL.

3. Business Hypothesis Requirement (Mandatory)

Each learner must define and validate at least **three business hypotheses** using data analysis.

Example hypotheses:

- Delivery delays negatively impact customer ratings
- High-value customers prefer specific payment methods
- Certain regions contribute disproportionately to revenue concentration

Hypotheses must be validated using statistical analysis or data visualization evidence.

4. Insight Differentiation Requirement

Learners must ensure that insights are not limited to descriptive reporting. Each submission must include:

- At least 3 analytical insights unique to their perspective
- At least 1 unexpected or non-obvious finding
- At least 1 data-driven business recommendation tied to measurable impact

5. Decision Simulation Requirement

Each learner must include a section titled:

“If I were the Business Head…”

This section must include:

- 3 prioritized strategic actions
- Expected business impact
- Trade-offs or risks
- Supporting data evidence

6. AI Usage Differentiation (Mandatory)

AI-assisted analytics tools must be used for:

- Generating alternative interpretations of findings
- Validating insights with counter-analysis prompts
- Refining business recommendations

All AI outputs must be critically evaluated and supported with dataset evidence.

9. Deliverables

Technical Deliverables

- GitHub Repository URL
- Data Acquisition Notebook (.ipynb)
- Excel Analysis Workbook (.xlsx)
- SQL Scripts (.sql)
- Statistical Analysis Report (.pdf)
- Power BI Dashboard (.pbix)

Business Deliverables

- Business Insight Report (.pdf)
- Executive Presentation (.pptx)
- AI Validation Documentation (.pdf)

10. Submission Structure

StudentID_Name_MilestoneProject/

```
├── README.md
    └── Project Overview, Business Problem Statement, Dataset Information, Methodology, Key Findings, Dashboard Screenshots, and Business Recommendations

├── 01_Data_Acquisition/
    └── StudentID_Name_DataAcquisition.ipynb

├── 02_Excel/
    └── StudentID_Name_ExcelAnalysis.xlsx

├── 03_SQL/
    ├── StudentID_Name_SQLScripts.sql
    ├── DatabaseSchema.pdf
    └── ER_Diagram.png

├── 04_Statistics/
    └── StudentID_Name_StatisticalAnalysis.pdf

├── 05_PowerBI/
    ├── StudentID_Name_PowerBI.pbix
    └── Dashboard_Screenshots/

├── 06_AI_Analytics/
    ├── AI_Usage_Log.pdf
    ├── Prompt_Documentation.pdf
    └── Validation_Notes.pdf

├── 07_Presentation/
    └── StudentID_Name_ExecutivePresentation.pptx

└── 08_Final_Report/
    ├── StudentID_Name_BusinessInsightReport.pdf
    ├── DataDictionary.pdf
    └── DataQualityAssessment.pdf
```

GitHub Submission Requirement

The final project must be submitted through a GitHub repository containing all project deliverables.

The repository must include a README.md file covering:

- Project Overview
- Business Problem Statement
- Dataset Information
- Tools and Technologies Used
- Project Methodology
- Key Findings and Insights
- Dashboard Screenshots
- Business Recommendations

Students must submit the GitHub Repository URL as part of the final project submission

11. File Naming Convention

Deliverable	File Name
Notebook	StudentID_Name_DataAcquisition.ipynb
Excel	StudentID_Name_ExcelAnalysis.xlsx
SQL	StudentID_Name_SQLScripts.sql
Statistics	StudentID_Name_StatisticalAnalysis.pdf
Power BI	StudentID_Name_PowerBI.pbix
Presentation	StudentID_Name_ExecutivePresentation.pptx
Report	StudentID_Name_BusinessInsightReport.pdf

12. Evaluation Rubric

Evaluation Criteria	Weightage
Business & Data Understanding	10%
Data Cleaning & Preparation	15%
Excel Analysis & Application	10%
Statistical Analysis & Interpretation	10%
MySQL Querying & Data Extraction	15%
Power BI Data Modeling & DAX	10%
Dashboard Design & Visualization	10%
Insights & Business Recommendations	10%
Presentation & Communication Skills	10%
Total	**100%**

13. Success Criteria

Projects will be evaluated based on:

- Business understanding
- Analytical thinking
- Data quality management
- Technical implementation
- Visualization effectiveness
- Data storytelling
- Actionable recommendations
- Executive communication

14. Executive Challenge

At the conclusion of the project, leadership expects you to answer the following question:

"If you were presenting directly to the CEO, what are the three most strategic actions the company should take next based on your analysis?"

Your answer must be supported by evidence derived from the data and aligned to measurable business outcomes. Recommendations should be prioritized based on business impact, feasibility, and supporting evidence from the analysis.

Leadership is interested not only in what happened, but also in why it happened and what actions should be taken next.